

Formal languages and automata 2023

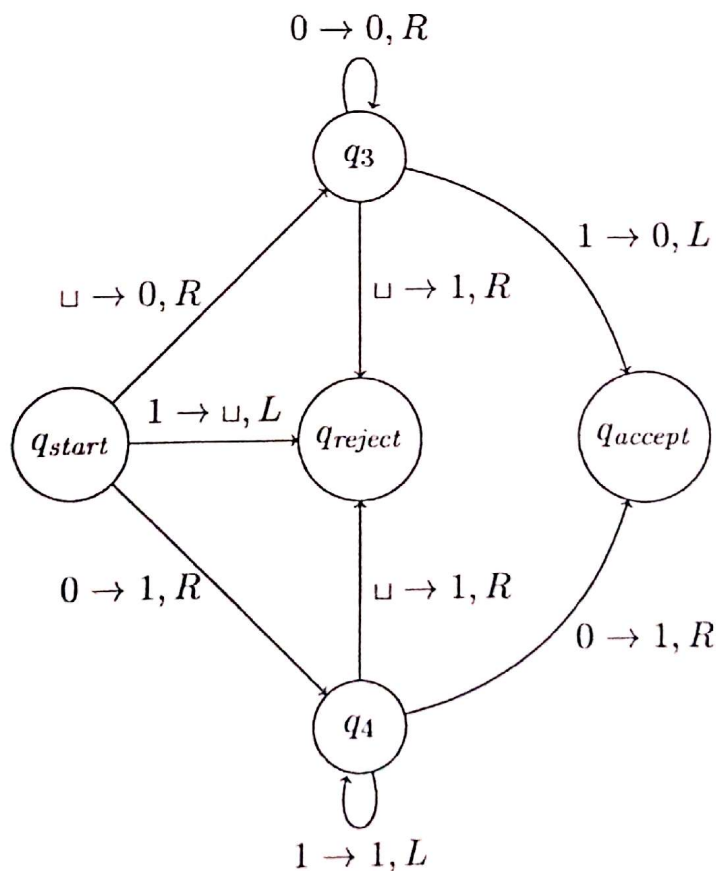
Test 2

15 September 2023

Total number of points: 40

Time: 45 mins

1. (10 points) Draw a diagram of a Turing machine that decides whether a string of 0s and 1s begins and starts with the same symbol (the empty string should be rejected).
2. (10 points) Encode the Turing machine depicted below in the alphabet $\{q, a, (,), 0, 1, ;\}$. Assume that the starting state has number 0, the accepting state has number 1, and the rejecting state has number 2.



3. (10 points) Prove that the set of all Turing machines is countable. If, in your proof, you want to use countability of some other set, you have to prove it, too.
4. (10 points) Prove that the set of the real numbers in the interval $[0, 1]$ is uncountable.